

# Learning Quantifiable Visual Explanations Without Ground-Truth

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## 1 Motivation



### Why Explanation Quality Matters

XAI is crucial for trustworthy and transparent AI, but evaluating explanations is challenging without ground-truth annotations.



### Limits of Existing Metrics

Perturbation-based metrics often disagree with human intuition—rewarding large or irrelevant regions instead of compact, sufficient ones.

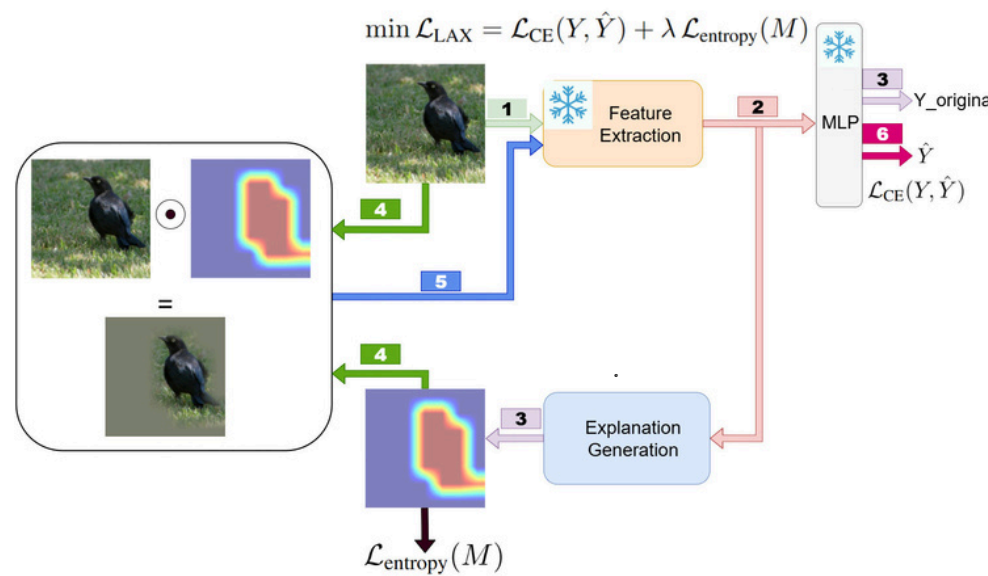


### Our Goal

We introduce a new metric, MSI, that balances sufficiency and minimality, and a novel method, LAX, that learns causal explanations without degrading model performance.

## 2 LAX (Learnable Adapter eXplanation)

We train a small adapter module on top of any frozen black-box model using a differentiable approximation of MSI.



### Key Benefits

- ✓ Adapter g is lightweight and trainable on any pre-trained model.
- ✓ Trained in a self-supervised way using MSI objective.
- ✓ Generates high-quality, causal explanations without ground-truth annotations or affecting the base model's accuracy.

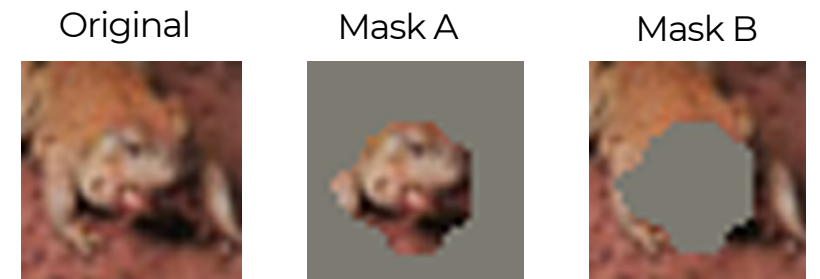
## 3 Metric: MSI

MSI evaluates whether an explanation is sufficient, compact, and robust to multiple valid explanations.

MSI addresses two key limitations of existing metrics.

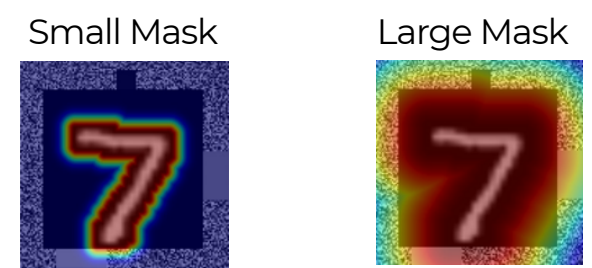
### (i) Multiple valid explanations

Different regions can be sufficient for the same prediction.

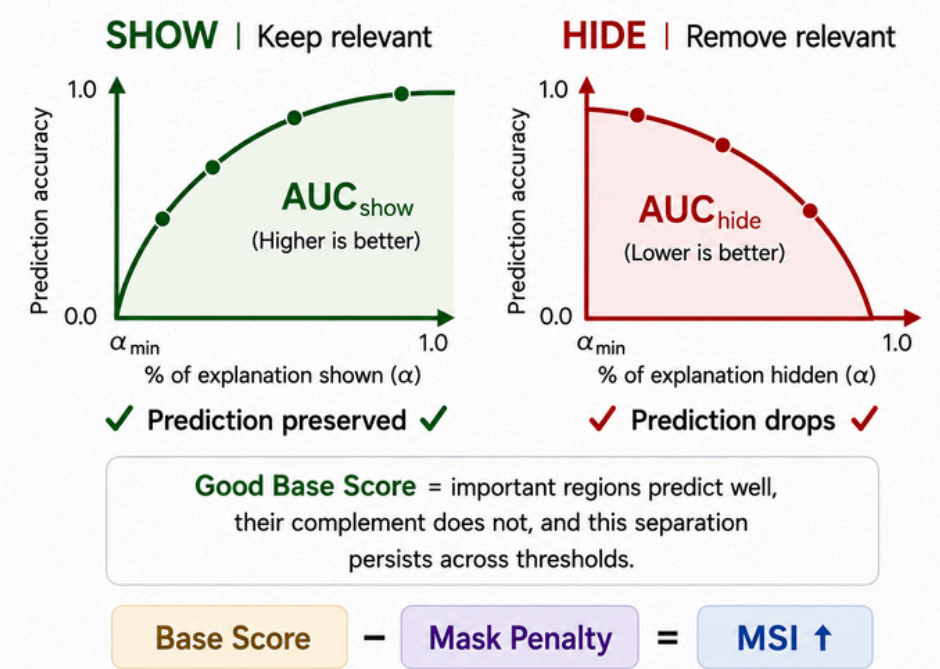


### (ii) Penalty for large masks

Oversized masks are penalized even if prediction is preserved



### MSI combines these ideas



Higher MSI ↑ means better explanations.

## 4 Quantitative Results

### Synthetic MNIST Dataset

| Method     | MoRF (Pixel Value) |            | MoRF (% Value) |            | Fidelity (Pixel Value) |               | Fidelity (% Value) |               | Ours                                   |             |           |
|------------|--------------------|------------|----------------|------------|------------------------|---------------|--------------------|---------------|----------------------------------------|-------------|-----------|
|            | Insertion ↑        | Deletion ↓ | Insertion ↑    | Deletion ↓ | Fid- (Ins.) ↓          | Fid+ (Del.) ↑ | Fid- (Ins.) ↓      | Fid+ (Del.) ↑ | BaseScore<br>( $\alpha_{\min} = 0.5$ ) | MaskPenalty | MSI Score |
| Grad-CAM   | 0.744              | 0.250      | 0.816          | 0.198      | 0.255                  | 0.745         | 0.183              | 0.798         | 0.509                                  | 0.418       | 0.091     |
| Grad-CAM++ | 0.712              | 0.268      | 0.804          | 0.204      | 0.287                  | 0.728         | 0.194              | 0.792         | 0.456                                  | 0.371       | 0.085     |
| Layer-CAM  | 0.716              | 0.266      | 0.805          | 0.203      | 0.282                  | 0.73          | 0.194              | 0.793         | 0.462                                  | 0.378       | 0.084     |
| KPCA-CAM   | 0.612              | 0.329      | 0.825          | 0.201      | 0.388                  | 0.666         | 0.175              | 0.794         | 0.316                                  | 0.187       | 0.128     |
| Finer-CAM  | 0.616              | 0.347      | 0.742          | 0.244      | 0.382                  | 0.649         | 0.256              | 0.751         | 0.301                                  | 0.318       | -0.027    |
| LAX        | 0.917              | 0.142      | 0.906          | 0.162      | 0.085                  | 0.851         | 0.092              | 0.833         | 0.787                                  | 0.205       | 0.582     |

### CIFAR-10

| Method     | MoRF (Pixel Value) |            | MoRF (% Value) |            | Fidelity (Pixel Value) |               | Fidelity (% Value) |               | Ours                                    |                   |                |
|------------|--------------------|------------|----------------|------------|------------------------|---------------|--------------------|---------------|-----------------------------------------|-------------------|----------------|
|            | Insertion ↑        | Deletion ↓ | Insertion ↑    | Deletion ↓ | Fid- (Ins.) ↓          | Fid+ (Del.) ↑ | Fid- (Ins.) ↓      | Fid+ (Del.) ↑ | Base Score<br>( $\alpha_{\min} = 0.4$ ) | Mask Penalty<br>↓ | MSI Score<br>↑ |
| Grad-CAM   | 0.637              | 0.384      | 0.668          | 0.339      | 0.301                  | 0.55          | 0.272              | 0.593         | 0.374                                   | 0.599             | -0.225         |
| Grad-CAM++ | 0.620              | 0.395      | 0.662          | 0.342      | 0.317                  | 0.541         | 0.278              | 0.590         | 0.353                                   | 0.576             | -0.222         |
| Layer-CAM  | 0.626              | 0.387      | 0.659          | 0.341      | 0.311                  | 0.547         | 0.281              | 0.591         | 0.364                                   | 0.596             | -0.232         |
| KPCA-CAM   | 0.371              | 0.674      | 0.596          | 0.447      | 0.566                  | 0.268         | 0.343              | 0.489         | -0.094                                  | 0.272             | -0.367         |
| Finer-CAM  | 0.621              | 0.411      | 0.674          | 0.352      | 0.316                  | 0.525         | 0.266              | 0.581         | 0.347                                   | 0.553             | -0.206         |
| LAX        | 0.739              | 0.581      | 0.735          | 0.423      | 0.207                  | 0.364         | 0.201              | 0.516         | 0.345                                   | 0.337             | 0.007          |

## 6 Conclusion



MSI provides a fair and reliable way to evaluate explanations without ground-truth.



LAX leverages MSI to learn compact, sufficient and semantically aligned explanations.



This work advances explainability towards more trustworthy AI systems.

## 7 Acknowledgements

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